

MEDIPRIORITY OUR AI-DRIVEN HEALTHCARE ECO-SYSTEM

Supporting Material

Medical Assistant

Home Questionnaire

Medical Assistant Chatbot

Assistant:

Hello! What is your name and age?

You:

My name is Sneh and I am 20 years old

Assistant:

What is the main symptom you are experiencing today?

You:

I am having cold and fever

Would you like to upload any images or videos for visual information?

Choose Files

No file chosen

Analyse

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Example analysis of visual agent

Medicare

Enter what you are going through:

i am going through a severe eye disease...

Upload a video file



Drag and drop file here

Limit 200MB per file • MP4, AVI, MOV, MKV, MPEG4

Browse files



videoplayback (1).mp4 1.8MB



Process

Processing complete!

Based on the visual input from the provided image, the following health indicators can be assessed:

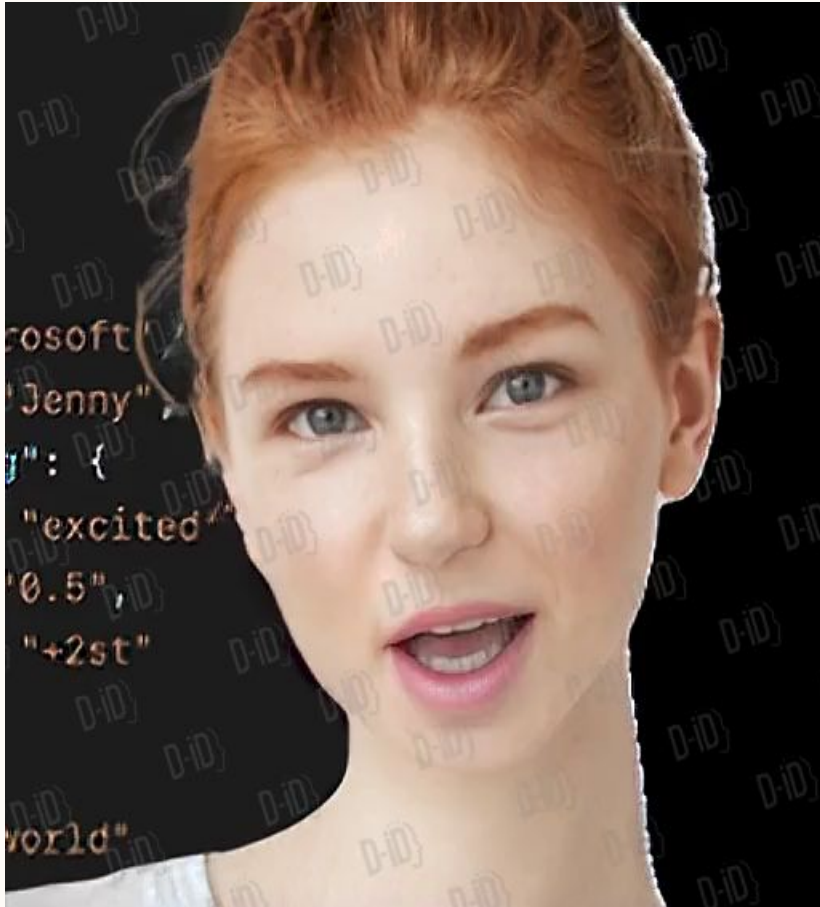
1. **Yellowish Skin (Jaundice):** The patient's skin appears slightly yellow, which may indicate jaundice. Jaundice is a condition characterized by the yellowing of the skin and whites of the eyes, and it is often a sign of liver problems or bile duct obstruction.
2. **Yellowing of the Sclera:** The sclera, which is the white part of the eye, also appears slightly yellow. This further supports the indication of jaundice.
3. **Dark Circles:** The patient has dark circles under the eyes, which may suggest sleep deprivation, stress, or chronic fatigue.
4. **Red or Bloodshot Eyes:** The patient's eyes appear slightly red or bloodshot. This could be a sign of infection, allergy, or irritation.
5. **Swollen Glands:** There is a visible swelling in the neck region, which may indicate swollen glands. Swollen glands can be a sign of infection or an immune response.

It's important to note that these observations are based solely on the visual input from the image and do not constitute a medical diagnosis. A proper medical evaluation is necessary to determine the exact cause of the symptoms and provide appropriate treatment.

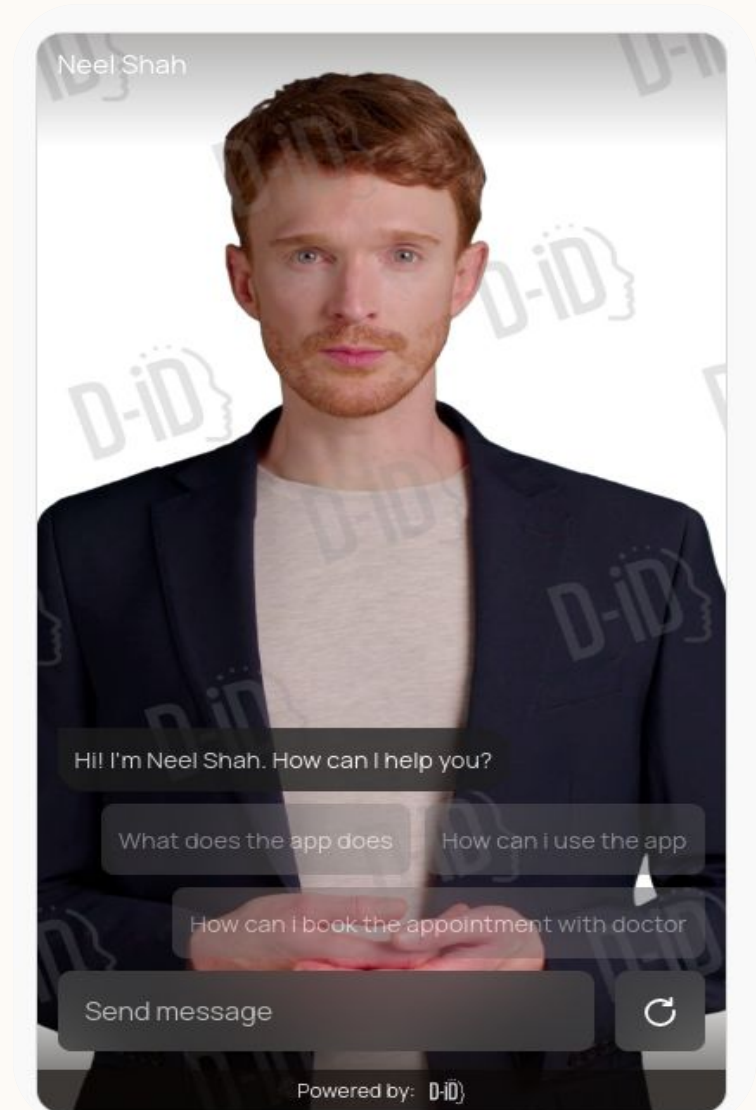
Your Score for urgency is:

2

Ai remedies recommender avatar



Live visual recommender avatar



Dynamic Patient slot booking

5

Creating doctor data

```
1 # Create some doctors
2 doctor1 = Doctor(1, "MD", 10, "New York", "Cardiologist", num_days=14)
3 doctor2 = Doctor(2, "PhD", 15, "New York", "Cardiologist", num_days=5)
4 doctor3 = Doctor(3, "MD", 8, "Los Angeles", "Neurologist", num_days=18)
5 matching_system.add_doctor(doctor1)
6 matching_system.add_doctor(doctor2)
7 matching_system.add_doctor(doctor3)
```

`/usr/local/lib/python3.10/dist-packages/huggingface_hub/utils/_token.py:89: UserWarning:`
The secret 'HF_TOKEN' does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (<https://huggingface.co/settings/token>)
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
warnings.warn(
1_Pooling/config.json: 0%| | 0.00/190 [00:00<?, ?B/s]
config.json: 0%| | 0.00/612 [00:00<?, ?B/s]
model.safetensors: 0%| | 0.00/90.9M [00:00<?, ?B/s]
tokenizer_config.json: 0%| | 0.00/350 [00:00<?, ?B/s]

Inserting patient data and adding them to the waiting list

```
1 # Create some patients
2 patient1 = Patient(1, "New York", "Stable", "video1.mp4", "Heart palpitations and shortness of breath", "Cardiologist", 5)
3 patient2 = Patient(2, "Los Angeles", "Critical", "video2.mp4", "Severe headache and confusion", "Neurologist", 7, is_emergency=True)
4 patient3 = Patient(3, "New York", "Stable", "video3.mp4", "Chest pain and dizziness", "Cardiologist", 6)
```

`+ Code` `+ Markdown`

```
1 # Add patients to waiting list
2 appointment_system.add_to_waiting_list(patient1)
3 appointment_system.add_to_waiting_list(patient2)
4 appointment_system.add_to_waiting_list(patient3)
```

```
1 # Process appointments
2 appointment_system.process_appointments(matching_system)
3
4 # Update waiting times and reprocess
5 appointment_system.update_waiting_times()
6 appointment_system.process_appointments(matching_system)
7
8 # Simulate an emergency patient arrival
9 emergency_patient = Patient(4, "New York", "Critical", "video4.mp4", "Severe chest pain and shortness of breath", "Ca
10 appointment_system.add_to_waiting_list(emergency_patient)
```

There is an availab

Dynamic Patient slot booking

Creating dummy data for real-life with 1000 patient and 12 doctors

```

1 # Generate dummy doctors
2 dummy_doctors = generate_dummyDoctors(numDoctors=12)
3
4 # Assuming the matching_system and its add_doctor method are already defined
5 for doctor in dummy_doctors:
6     matching_system.add_doctor(doctor)
7
8 # Print a few examples to verify
9 for doctor in dummy_doctors:
10     print(f"Doctor {doctor.id}: {doctor.degree}, {doctor.experience} years, {doctor.location}, {doctor.specialization}")
11

```

Doctor 1: DO, 30 years, Chicago, Cardiologist
 Doctor 2: DO, 11 years, Philadelphia, Neurologist
 Doctor 3: PhD, 13 years, Philadelphia, Oncologist
 Doctor 4: PhD, 18 years, Houston, Pediatrician
 Doctor 5: MBBS, 19 years, Houston, Dermatologist
 Doctor 6: MBBS, 14 years, New York, Orthopedist
 Doctor 7: DO, 19 years, New York, Gynecologist
 Doctor 8: DO, 7 years, Los Angeles, Urologist
 Doctor 9: DO, 24 years, Houston, Psychiatrist
 Doctor 10: MD, 8 years, Chicago, Endocrinologist
 Doctor 11: PhD, 14 years, Los Angeles, Orthopedist
 Doctor 12: MD, 13 years, Chicago, Pediatrician

```

1 # Generate 1000 dummy patients
2 dummy_patients = generate_dummyPatients(1000)
3
4 # Print a few examples to verify
5 for patient in dummy_patients[:10]:
6     print(f"Patient {patient.id}: {patient.location}, {patient.health_status}, {patient.required_doctor_type}, "
7           f"Emergency: {patient.is_emergency}, Score: {patient.score}, Symptoms: {patient.text_description}")
8
9 print("Length of the dummy patients: ", len(dummy_patients))

```

Patient 1: San Antonio, Serious, Orthopedist, Emergency: False, Score: 7.0, Symptoms: Back pain and Osteoporosis
 Patient 2: Los Angeles, Critical, Pediatrician, Emergency: True, Score: 2.0, Symptoms: Sore throat and Rash
 Patient 3: Philadelphia, Stable, Gynecologist, Emergency: False, Score: 7.0, Symptoms: Vaginal discharge and Breast lump
 Patient 4: San Diego, Critical, Psychiatrist, Emergency: True, Score: 7.0, Symptoms: Insomnia and Mood swings
 Patient 5: Houston, Critical, Psychiatrist, Emergency: True, Score: 6.0, Symptoms: Anxiety and Insomnia
 Patient 6: San Antonio, Critical, Endocrinologist, Emergency: True, Score: 4.0, Symptoms: Weight management issues and
 Patient 7: New York, Stable, Urologist, Emergency: False, Score: 6.0, Symptoms: Erectile dysfunction and Blood in urine
 Patient 8: San Diego, Stable, Urologist, Emergency: False, Score: 3.0, Symptoms: Blood in urine and Erectile dysfunction
 Patient 9: San Diego, Moderate, Pediatrician, Emergency: False, Score: 2.0, Symptoms: Fever and Sore throat
 Patient 10: Houston, Critical, Gynecologist, Emergency: True, Score: 5.0, Symptoms: Breast lumps and Abnormal bleeding
 Length of the dummy patients: 1000

Dynamic Patient slot booking

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Allocating time to the patients

```
Scheduled appointment for patient 911 with doctor 10 at 2024-08-23 09:00:00
Waiting time: 17 days
Scheduled appointment for patient 115 with doctor 11 at 2024-08-16 10:00:00
Waiting time: 10 days
Scheduled appointment for patient 411 with doctor 2 at 2024-08-20 10:00:00
Waiting time: 14 days
Scheduled appointment for patient 959 with doctor 3 at 2024-08-06 12:00:00
Waiting time: 0 days
Scheduled appointment for patient 198 with doctor 11 at 2024-08-16 11:00:00
Waiting time: 10 days
Scheduled appointment for patient 963 with doctor 11 at 2024-08-16 12:00:00
Waiting time: 10 days
Scheduled appointment for patient 636 with doctor 1 at 2024-08-09 14:00:00
Waiting time: 3 days
Scheduled appointment for patient 668 with doctor 9 at 2024-08-21 15:00:00
Waiting time: 15 days
Scheduled appointment for patient 988 with doctor 3 at 2024-08-22 16:00:00
Waiting time: 17 days
Scheduled appointment for patient 125 with doctor 4 at 2024-08-09 09:00:00
Waiting time: 3 days
Scheduled appointment for patient 926 with doctor 1 at 2024-08-09 15:00:00
Waiting time: 3 days
Scheduled appointment for patient 202 with doctor 3 at 2024-08-23 09:00:00
Waiting time: 17 days
Scheduled appointment for patient 463 with doctor 3 at 2024-08-23 10:00:00
Waiting time: 17 days
Scheduled appointment for patient 228 with doctor 12 at 2024-08-15 13:00:00
Waiting time: 9 days
Scheduled appointment for patient 802 with doctor 3 at 2024-08-23 11:00:00
Waiting time: 17 days
```

Doctor Meet Sneh Shah

Meeting started: 6/28/2024, 11:01:41 AM

Duration: 10 minutes

Participants: Sneh Shah, Harishon Rich

00:00 Harishon Rich: Sneh, how are you feeling today?

00:05 Sneh Shah:Hi, Doctor. I still have a high fever, cold, and cough. It's been like this for the

00:35 Harishon Rich:I see. Based on the symptoms you've described-high fever, cold, and cough-it sounds like you have a viral infection. These symptoms usually resolve on their own with proper rest and medication. However, it's important to monitor

01:00 Sneh Shah:That makes sense. What can I do to feel better?

01:35 Harishon Rich:Continue taking Paracetamol for the fever. It's good to hear that it has been effective for you. Take it every 6 hours as needed for fever, Cetirizine 10mg before bed for your cold symptoms, and Guaifenesin

02:45 Harishon Rich:Make sure to drink plenty of fluids and get plenty of rest.

03:10 Sneh Shah:Got it. How long should I continue taking these medications?

04:00 Harishon Rich:Take the medications for the next three days. If your symptoms persist or worsen, please reach out to me. Make sure you are recovering well.

05:01 Sneh Shah:Okay, I will do that. Thank you, Doctor.

05:60 Harishon Rich:You're welcome, Sneh. Take care and get well soon. Don't hesitate to reach out if you need further assistance.

```
{
  "recovery_recommendations": [
    "Continue taking Paracetamol for the fever. It's good to hear that it has been effective for you.",
    "Take Cetirizine 10mg before bed for your cold symptoms.",
    "Take Guaifenesin 200mg after breakfast and after dinner for the cough.",
    "Make sure to drink plenty of fluids.",
    "Get plenty of rest.",
    "Monitor the fever and ensure it doesn't get too high."
  ],
  "medication_details": [
    {
      "name": "Paracetamol",
      "dose": "500mg",
      "start_date": "2024-06-28",
      "end_date": "2024-06-30",
      "timings": [
        "8:00 AM",
        "2:00 PM",
        "8:00 PM",
        "10:00 PM"
      ]
    },
    {
      "name": "Cetirizine",
      "dose": "10mg",
      "start_date": "2024-06-28",
      "end_date": "2024-06-30",
      "timings": [
        "10:00 PM"
      ]
    },
    {
      "name": "Guaifenesin",
      "dose": "200mg",
      "start_date": "2024-06-28",
      "end_date": "2024-06-30",
      "timings": [
        "8:00 AM",
        "8:00 PM"
      ]
    }
  ],
  "follow_up_date": "2024-07-03"
}
```

Creating google calendar reminders for medicines and appointments

Please visit this URL to authorize this application: https://accounts.google.com/authorize?client_id=123456789012345678901&redirect_uri=https://example.com/callback&scope=https://www.googleapis.com/auth/calendar

adding event for Paracetamol at 2024-06-28T20:00:00 for 3 days
Event created: <https://www.google.com/calendar/event?eid=b3VtanV10>

adding event for Paracetamol at 2024-06-28T20:00:00 for 3 days
Event created: <https://www.google.com/calendar/event?eid=bXY3bHFsb>

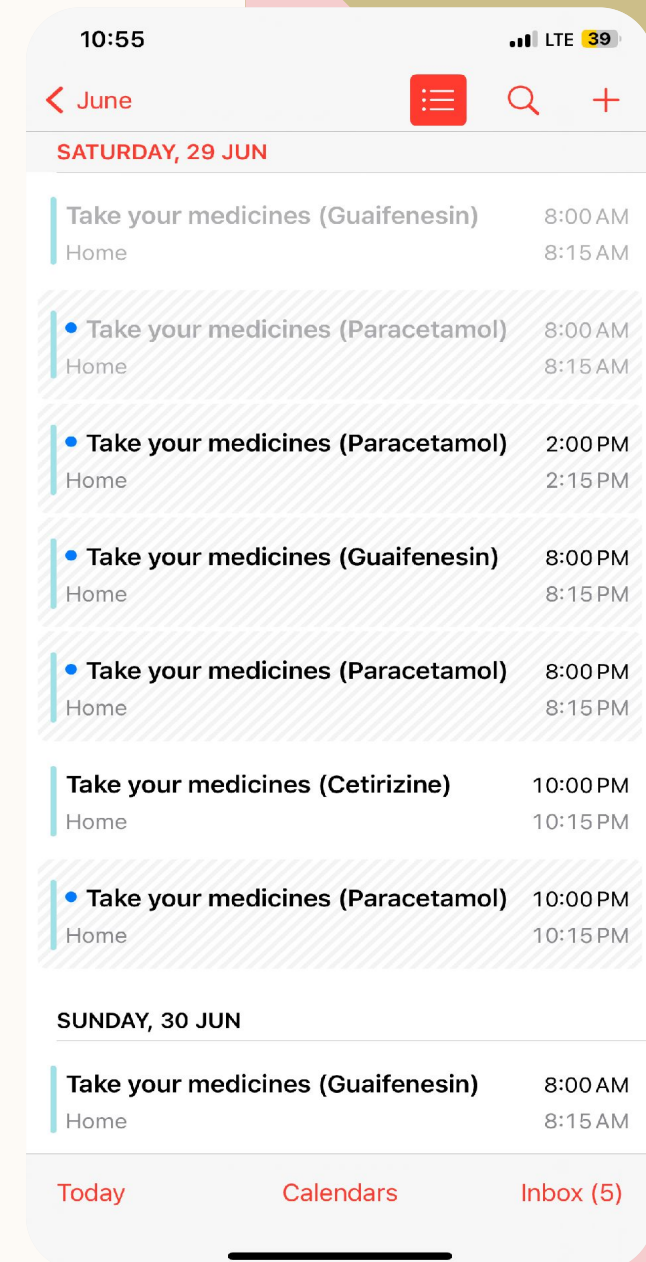
adding event for Paracetamol at 2024-06-28T20:00:00 for 3 days
Event created: <https://www.google.com/calendar/event?eid=MjJkMHRrd>

adding event for Paracetamol at 2024-06-28T20:00:00 for 3 days
Event created: <https://www.google.com/calendar/event?eid=a3Rob3Bqa>

adding event for Cetirizine at 2024-06-28T20:00:00 for 3 days
Event created: <https://www.google.com/calendar/event?eid=MjVrNW43M>

adding event for Guaifenesin at 2024-06-28T20:00:00 for 3 days
Event created: <https://www.google.com/calendar/event?eid=MjZ0Y2NuM>

adding event for Guaifenesin at 2024-06-28T20:00:00 for 3 days
Event created: <https://www.google.com/calendar/event?eid=a2sybTI3d>



Analysis of Prescription

Ma. Lh. ~ Ma in Date 12-6-2014
56/F
Concern: Neck mass - noted 2 wks ago
no pain
Physical Exam Findings (Neck):

⇒ Thyroid Mass Right Lobe

① ~ 2cm
worse w/ swallowing
feels solid/firm
no tenderness
no redness
Options for further Diagnostic Test:
① Biopsy, needle Benefit Risk Cost Avail
90% Pain TT Recov
Infect, low TT Recov
no/N ammda 2 Ultrasound Indirect fauten

Thyroid Disorder
Functional Structural / Masses < Cancer
Hormonal
Hypo + No symptom = Forgive = Normal
Hypertension RR

Lic. # 44609

Name: Lolita Alvarez
Address: Bagongilog, Pasig City
Age: 39 Sex: F Date: 2-5-13

Rx

FeSO₄ tab # 30

Sig: A.D.

Ascorbic Acid # 30
300mg tab

Sig: Once a day

Physician's Sig. J. Alvarez
Lic. No. 12345
PTR No. 1234567
S2 No.

System Instructions

Ma. Lh. ~ Ma in Date 12-6-2014
56/F
Concern: Neck mass - noted 2 wks ago
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Physical Exam Findings (Neck):

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Thyroid Disorder
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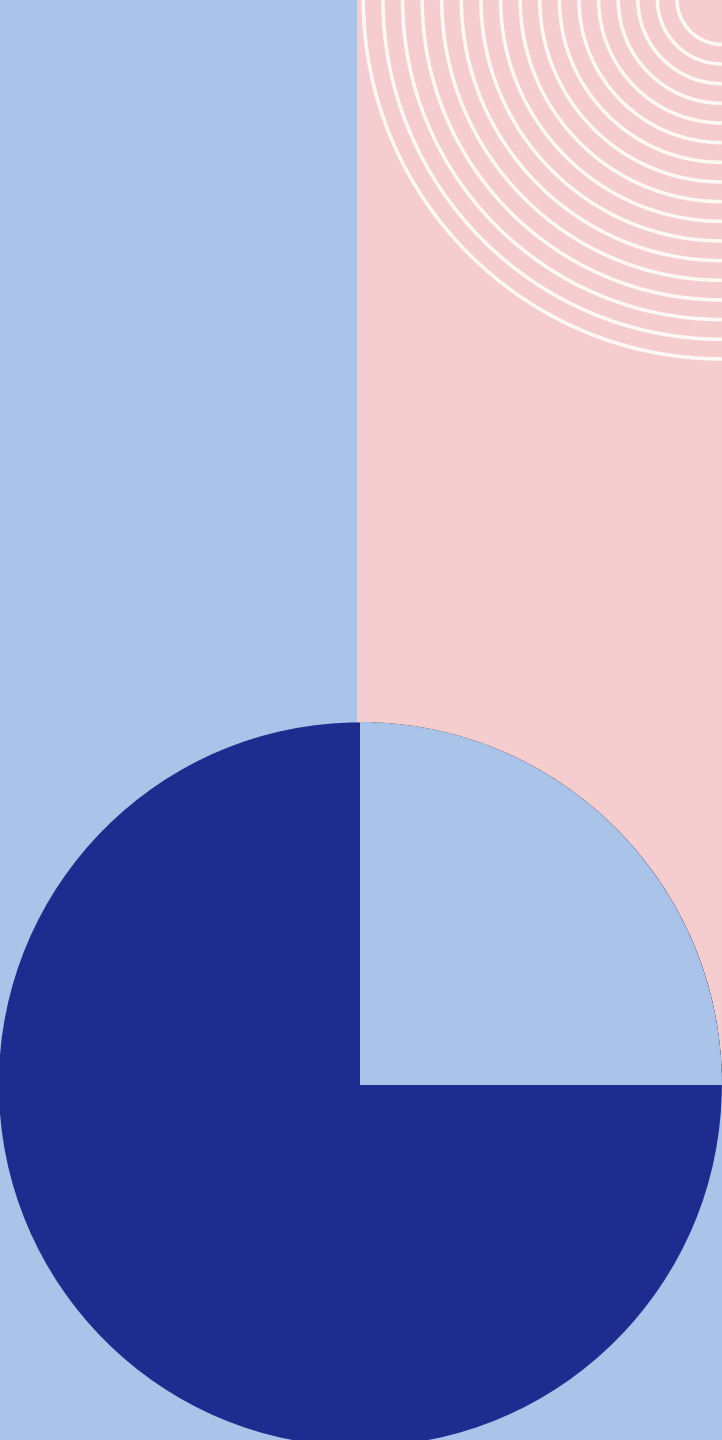
User

Extract me the text written in the below image

Model 6.5s

Ma. Lh. ~ Ma in 12-6-14
56/F Date 20
Concern: Neck mass - noted 2 wks ago
no pain
Physical Exam Findings (Neck):

- Thyroid Mass Right Lobe R ~ 2cm Left
- CANCER NOT CANCER
- worse w/ swallowing - = Need for FNA!
- feels solid / firm
- no tenderness
- Options for further Diagnostic Test:
- P: Bx:
- An: no redness ① Biopsy, needle Benefit Risk Cost Avail
- // Durant Pain TT Recov
- 90% Bleed/Rx
- Infect, low TT Recov
- no/N ammda 2. Ultrasound Indirect fauten



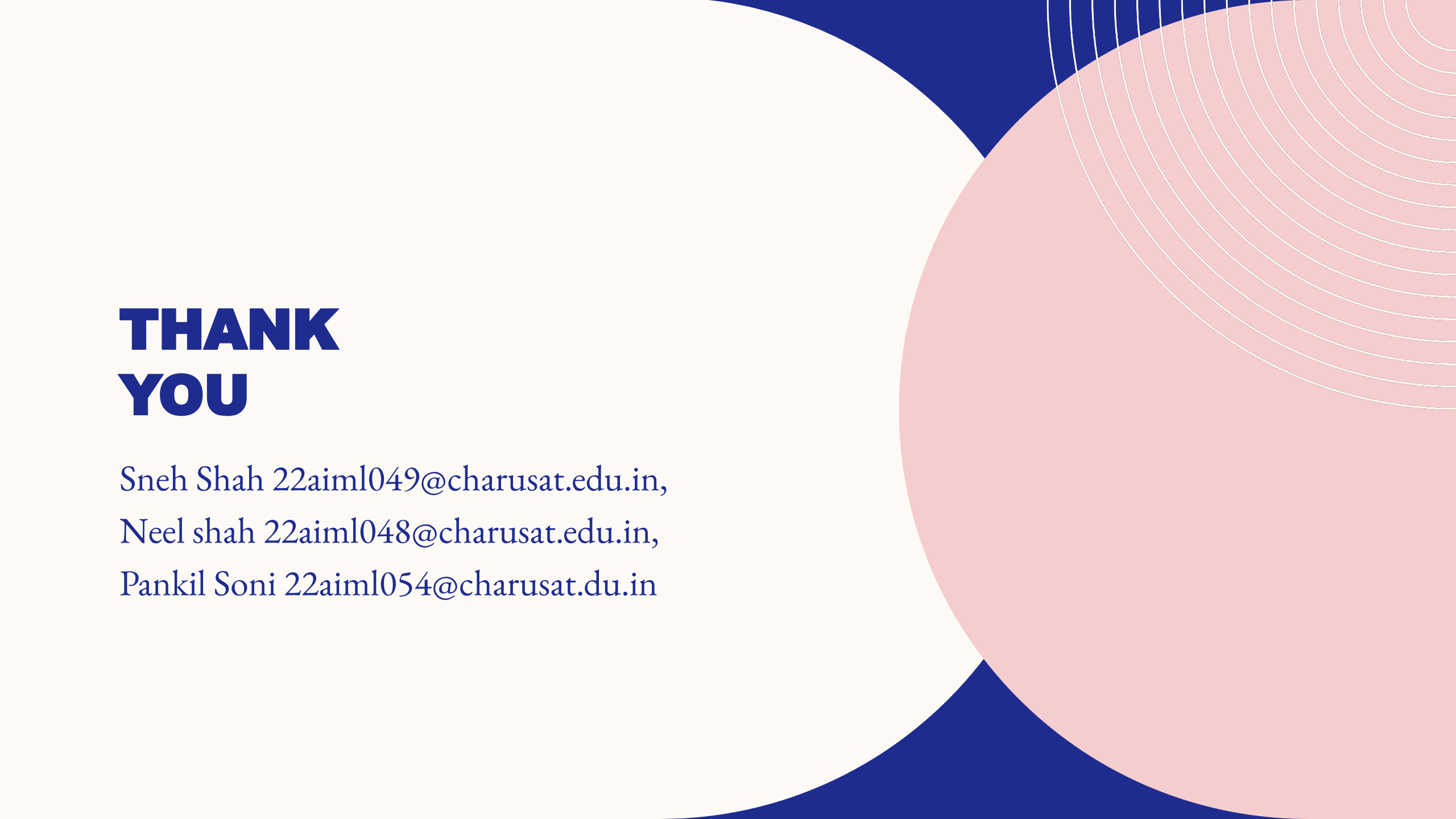
FUTURE SCOPE

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- Add more interpretability and fairness to all of the AI models.
- Experiment with custom designed models instead of only depending on the LLMs.
- Integrate all the existing models with Front-end and back-end and make entire system more scalable.

NEW FEATURES THAT WE ARE PLANNING

- Predictive Health AI: Forecast potential health issues before symptoms appear.
- Graph-Integrated Care: Personalized treatment based on genetic/similar profiles.
- AI Emotional Support: Advanced chatbots for mental health assistance.



THANK YOU

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